KPPSC PMS Geography Paper-I — Q.1: Internal Structure of Earth & Seismic Discontinuities

KPPSC PMS Master Examination Notes — Paper-I (Physical & General Geography)

Target Examination	KPPSC PMS (Provincial Management Service) — Khyber Pakhtunkhwa
Paper & Subject	Geography Paper-I (Physical & General Geography — 100 Marks)
EXAM QUESTIONS LINKED:	• **2025 Q.1:** "Describe the internal structure of the Earth and explain how seismic wave behavior provides evidence for the layering of the Earth's interior. Evaluate the scientific reliability of seismic methods in understanding deep Earth processes. (20 Marks)" • **2016 Q.3:** "What are the different layers of the Earth? Relate the three types of plate collisions associated with seismicity, volcanism, and mountain building. (20 Marks)" • **2008 Q.5:** "Define the internal structure of the Earth from Asthenosphere to inner core. Give reasons why the core is composed of iron. (20 Marks)" • **2008 Q.2:** "Describe the mechanism and types of seismic waves. Define the causes of earthquakes and write an account of earthquakes in Pakistan. (20 Marks)"
Target Syllabus	Section 2: Landforms — Internal Structure of the Earth & Seismology
Marks & Allocation	20 Marks | Strict 35-Minute Descriptive Writing Limit
Core Theoretical Thesis	The Earth's concentric internal architecture is deciphered indirectly via the refraction, reflection, and shadow-zone mechanics of body seismic waves (P and S waves). The abrupt velocity shifts across first-order discontinuities (Conrad, Mohorovičić, Repetti, Gutenberg, Lehmann) demarcate sharp rheological and compositional boundaries—transitioning from brittle silicate crust to plastic peridotite mantle, liquid outer iron-nickel core, and crystalline solid inner core.

🎯 Exam Questions Linked

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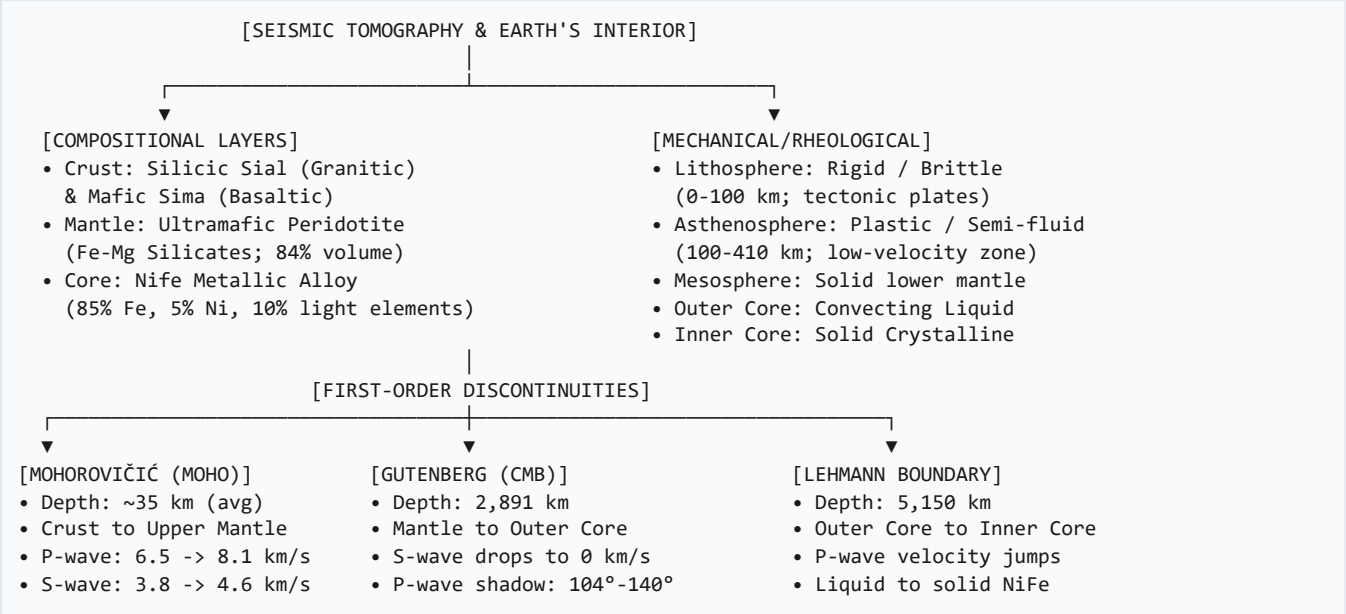
SEISMIC PROBING OF THE INTERIOR: Direct borehole drilling (e.g., Kola Superdeep Borehole at 12.26 km) penetrates less than 0.2% of Earth's 6,371 km radius. Geoscientists rely on body seismic waves generated by high-magnitude earthquakes as natural tomographic probes.

P-WAVES VS. S-WAVES DICHOTOMY: Primary (P) waves are compressional longitudinal waves (propagating through solids, liquids, and gases), whose velocity varies with incompressibility and density (Vp = sqrt[(K + 4/3*mu)/rho]). Secondary (S) waves are transverse shear waves that cannot propagate through liquids (shear modulus mu = 0).

THE TWO FIRST-ORDER DISCONTINUITIES: The Mohorovičić Discontinuity (Moho, ~30–70 km under continents, 5–10 km under oceans) marks the crust-mantle chemical boundary where P-wave velocity jumps from ~6.5 to ~8.1 km/s. The Gutenberg Discontinuity (2,891 km) marks the core-mantle boundary where S-waves abruptly terminate and P-waves refract sharply downward, creating the diagnostic seismic shadow zones.

CORE COMPOSITION & GEODYNAMO: The core's high density (~10–13 g/cm^3) and magnetic properties confirm an iron-nickel (Nife) composition with trace light elements (silicon, sulfur, oxygen). Convection currents in the liquid outer core drive the geodynamo that generates Earth's geomagnetic field.

1. Conceptual Mind Map & Thematic Roadmap



2. Analytical Dimensions & Geomorphic/Systemic Matrix

Internal Boundary / Layer	Depth & Physical State	Seismic Behavior & Petrographic Evidence
Continental & Oceanic Crust	0–35 km (avg); 5–10 km oceanic, 30–70 km continental (solid)	P-waves travel at 6.0–6.8 km/s; granitic Sial (2.7 g/cm^3) transitions across Conrad discontinuity to basaltic Sima (3.0 g/cm^3).
Mohorovičić Discontinuity (Moho)	Depth: 30–70 km under continents; 7–10 km under oceanic basins	First-order seismic jump discovered by Andrija Mohorovičić (1909); P-wave speed leaps from ~6.5 to 8.1 km/s as rock density shifts to ultramafic peridotite.
Asthenosphere (Low-Velocity Zone)	100–410 km (partial melt: 1–5% liquid; ductile plastic)	Pronounced drop in P and S wave velocities due to rheological softening; enables isostatic adjustment and lithospheric plate motion.
Mesosphere (Lower Mantle)	660–2,891 km (solid high-density peridotite/bridgmanite)	Continuous velocity increase under extreme confining pressure; P-waves reach ~13.7 km/s; terminates at D'' layer above core boundary.
Gutenberg Discontinuity (CMB)	Depth: 2,891 km (mantle-to-core boundary)	Identified by Beno Gutenberg (1914); S-waves vanish completely (shear modulus = 0); P-wave velocity plummets from 13.7 to 8.1 km/s due to liquid iron state.
Outer Core	2,891–5,150 km (molten/liquid iron-nickel alloy)	True liquid medium (density 9.9–12.2 g/cm^3); vigorous thermal convection sustains geodynamo; creates global S-wave shadow zone beyond 104° epicentral distance.
Lehmann Discontinuity & Inner Core	5,150–6,371 km (solid crystalline iron-nickel sphere)	Discovered by Inge Lehmann (1936); weak P-wave reflections confirm solid state; P-wave velocity jumps from 10.3 to 11.2 km/s due to 3.6 million atm pressure.

3. 20-Mark Master Model Essay

Introduction: The Subterranean Frontier and Seismological Remote Sensing

Human direct empirical observation of the Earth's interior remains confined to the outermost skin of the lithosphere. The deepest artificial excavation in human history—the Kola Superdeep Borehole in Russia—penetrated a mere 12.26 kilometers, representing less than 0.2 percent of the planet's 6,371-kilometer mean radius. Consequently, understanding the deep interior relies upon geophysical remote sensing, predominantly the analysis of body seismic waves produced by high-energy tectonic earthquakes. By monitoring how Primary (P) compressional waves and Secondary (S) shear

waves refract, reflect, accelerate, and extinguish as they traverse the Earth, seismologists reconstruct the concentric compositional and mechanical layers of our planet. This seismological tomography reveals a dynamic, highly differentiated terrestrial engine consisting of an outer brittle crust, a silicate plastic mantle, a convecting liquid metallic outer core, and a super-pressurized, crystalline solid inner core.

Mechanics of Body Seismic Waves: The Diagnostic Probes of Earth's Interior

Seismic waves generated at earthquake foci divide fundamentally into surface waves (Love and Rayleigh waves) and body waves (Primary 'P' and Secondary 'S' waves). For deep interior profiling, body waves serve as the definitive diagnostic tools. Primary waves are longitudinal, compressional stress waves that vibrate rock particles parallel to the direction of wave propagation. Mathematically, P-wave velocity is governed by the formula Vp = sqrt[(K + 4/3*mu)/rho], where K represents the bulk modulus (incompressibility), mu is the shear modulus (rigidity), and rho is the density. Because both solids, liquids, and gases resist compression (K > 0), P-waves can propagate through all states of matter, though their velocity drops sharply when passing from solid to liquid due to the collapse of shear rigidity. In contrast, Secondary waves are transverse shear waves vibrating perpendicular to propagation, with velocity expressed as Vs = sqrt(mu/rho). Crucially, liquids possess zero resistance to shear stress (mu = 0), rendering S-waves physically incapable of traversing molten or liquid media. The differential travel times, refraction trajectories (governed by Snell’s Law in spherical coordinates), and shadow zones of these two wave types provide precise, mathematical evidence of deep structural boundaries.

The Concentric Interior: Crust, Mantle, and Rheological Stratification

The Earth exhibits a twofold classification: a chemical/compositional division (Crust, Mantle, Core) and a mechanical/rheological division (Lithosphere, Asthenosphere, Mesosphere, Outer Core, Inner Core). The outermost crust comprises two distinct petrographic domains: the continental crust, averaging 35 km in thickness (reaching 70 km beneath orogenic belts like the Himalayas), composed of low-density (2.7 g/cm^3) granitic-andesitic rocks rich in silica and aluminum (Sial); and the oceanic crust, 5 to 10 km thick, consisting of dense (3.0 g/cm^3) basaltic-gabbroic rocks rich in silica and magnesium (Sima). Beneath the crust lies the mantle, extending to a depth of 2,891 km and constituting roughly 84 percent of Earth's volume. Composed primarily of ultramafic peridotite (iron- and magnesium-rich silicates), the mantle is subdivided mechanically. The uppermost solid mantle combines with the crust to form the rigid, brittle Lithosphere (0–100 km). Directly beneath lies the Asthenosphere (100–410 km), a zone of high geothermal heat where 1 to 5 percent partial melting occurs. This creates the seismic 'Low-Velocity Zone' (LVZ), imparting plastic, ductile rheology upon which lithospheric plates glide. Beneath the 660-km transition zone lies the solid Mesosphere, where intense lithostatic pressure prevents rock melting despite temperatures exceeding 2,500°C.

First-Order Seismic Discontinuities and Shadow Zone Geometry

The internal layering is demarcated by first-order seismic discontinuities where wave velocities alter abruptly. The Mohorovičić Discontinuity ('Moho'), discovered in 1909 by Croatian seismologist Andrija Mohorovičić, separates the crust from the upper mantle. At the Moho, P-wave velocities leap abruptly from ~6.5 km/s to 8.1 km/s, reflecting the compositional transition to dense peridotite. In 1914, German geophysicist Beno Gutenberg identified the Core-Mantle Boundary (CMB) at 2,891 km depth. At the Gutenberg Discontinuity, S-waves terminate completely, demonstrating that the outer core is liquid, while P-waves suffer a catastrophic velocity collapse from ~13.7 km/s to ~8.1 km/s. This extreme refraction bends P-waves inward, producing a distinct 'P-wave Shadow Zone' between 104° and 140° angular distance from the earthquake epicenter. Furthermore, because S-waves cannot traverse the liquid core at all, an expansive 'S-wave Shadow Zone' extends across the entire hemisphere beyond 104° to 180°. In 1936, Danish seismologist Inge Lehmann discovered faint P-wave reflections within the shadow zone, proving the existence of a solid Inner Core at a depth of 5,150 km, where P-wave velocity abruptly accelerates back to 11.2 km/s.

The Core Composition, Iron Geochemistry, and the Geodynamo Mechanism

Multiple independent lines of scientific reasoning validate that the Earth's core is predominantly composed of an iron-nickel alloy (Nife). First, global gravitational measurements and orbital dynamics determine Earth's mean density to be 5.515 g/cm^3, whereas superficial crustal rocks average only 2.7 to 3.0 g/cm^3, mathematically demanding a hyper-dense core (10.0 to 13.0 g/cm^3). Second, geochemical analyses of primitive chondritic meteorites—representing the pristine primordial material from which the solar system accreted—demonstrate high metallic iron and nickel concentrations that must have segregated into the planetary center during Earth's early differentiation (the 'iron catastrophe'). Third, laboratory diamond-anvil cell experiments confirm that under pressures exceeding 3.3 to 3.6 million atmospheres and

temperatures of 5,000°C to 6,000°C, iron assumes a hexagonal close-packed (hcp) solid crystal structure matching the inner core's seismic velocity. The liquid outer core's thermal and compositional convection, combined with Coriolis forces generated by planetary rotation, acts as a self-exciting geodynamo. This vigorous circulation of molten conductive iron sustains the geomagnetic dipolar field, shielding the biosphere from lethal solar cosmic radiation.

Conclusion: The Unified Seismotectonic Architecture of the Planet

In evaluative summation, the internal structure of the Earth represents a masterwork of thermal differentiation and geochemical stratification. Seismic wave mechanics have transformed our understanding of deep Earth from mythological speculation into rigorous quantitative geophysics. The behavior of P and S waves across the Moho, Gutenberg, and Lehmann discontinuities provides unambiguous, replicable evidence of solid, plastic, liquid, and crystalline zones. Far from being a static interior, the deep Earth operates as a coupled thermodynamic heat engine: mantle plumes rise from the D'' thermal boundary layer above the Gutenberg discontinuity, subducting tectonic slabs sink to the core-mantle interface, and outer-core convection generates the protective magnetosphere. Mastery of these seismic principles is foundational to explaining surface volcanism, orogenesis, earthquake distribution in active belts like Pakistan's Chaman and MBT fault lines, and the overarching paradigm of plate tectonics.

4. Scholarly Perspectives & Geomorphic Axioms

Scholar / Geophysicist	Theoretical Contribution / Concept	Exact Quotable Academic Thesis / Axiom
Andrija Mohorovičić (1909)	Discovery of Crust-Mantle Boundary	*"The sudden refraction and velocity jump of seismic waves at shallow depth proves an abrupt compositional transition from crustal silicates to dense ultramafic mantle rock."*
Beno Gutenberg (1914)	Core-Mantle Boundary & S-Wave Extinction	*"The complete cessation of transverse shear waves at 2,891 km constitutes irrefutable mechanical proof that the Earth's outer core behaves as a true hydrodynamic liquid."*
Inge Lehmann (1936)	Discovery of the Solid Inner Core	*"Weak P-wave arrivals refracted into the core shadow zone demonstrate the existence of a distinct, highly rigid inner core suspended within the molten outer envelope."*
Arthur Holmes (1944)	Principles of Physical Geology	*"The Earth's internal thermal engine, sustained by radiogenic decay and primordial heat, drives convection currents in the plastic substratum that dictate all surface tectonic drama."*
Francis Birch (1952)	Elasticity and Constitution of the Earth's Interior	*"High-pressure shockwave physics confirms that the core consists primarily of metallic iron alloyed with nickel and approximately ten percent light elements like silicon, sulfur, or oxygen."*